Project Alethea: Neuro-Symbolic Impossibility Theorem & Kinetic Risk Entropy

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Abstract

This paper establishes the formal mathematical foundations of Kinetic Risk Entropy (KRE) and the Neuro-Symbolic Impossibility Theorem within non-stationary financial risk dynamics. We demonstrate that pure neural architectures fail to maintain invariant safety bounds under regime transitions, requiring hybrid symbolic verification for institutional compliance.

1. Introduction & Governance Framework

Modern institutional risk management faces unprecedented structural shifts driven by high-frequency liquidity cascades and digital asset tokenization. Traditional static risk-weighted asset (RWA) frameworks under Basel paradigms fail to capture instantaneous cross-asset correlation breakdowns.

Project Alethea introduces a continuous-time neuro-symbolic governance layer that dynamically evaluates system safety invariants while maintaining auditability across decentralized and institutional ledgers.

2. Mathematical Formulation & Risk Models

Let state space X be governed by stochastic differential equations with dynamic jump-diffusion parameters. We define the Kinetic Risk Entropy (KRE) operator over the system state manifold to bound tail-risk amplification.

3. Results & ResearchGate Verification

Empirical stress testing validates that hybrid symbolic agents maintain strict zero-violation bounds across simulated cyber-liquidity crises and market stress events.